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24 June 1954
SUPERSEDING
PA-PD-229
26 October 1953

MILITARY SPECIFICATION

COATED CELLOPHANE FILM FOR INCREMENT CHARGE CONTAINERS

1. SCOPE

1.1 This specification covers one grade of sheeting for use in increment charge containers for mortar ammunition.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406 -Plastics, Organic: General Specifications, Test Methods
UU-P-31 -Paper; General Specifications and Methods of Testing

MILITARY

MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition
MIL-P-10025 -Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for

DRAWINGS

ORDNANCE CORPS

DP-44282 -120° Vacuum Stability Test Apparatus

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 The material shall consist of sheeting (which shall have the prior approval of the Office, Chief of Ordnance) such as regenerated cellulose impregnated with a plasticizing agent (the type and quantity of which shall have the prior approval of the Office, Chief of Ordnance) and coated with cellulose nitrate or a copolymer of vinylidene chloride and vinyl chloride in which the vinylidene chloride shall constitute 81 to 85% of the weight of the constituent chlorides. The coating shall be on both sides of the regenerated cellulose, and shall be applied in such a manner as to permit heat sealing of the sheets. The finished sheeting shall be essentially free from the volatile solvents used in coating.

3.2 Color.-The sheeting shall be essentially colorless.

3.3 Bursting strength.-The sheeting shall have a bursting strength of not less than 38 points per 0.001 inch thickness of regenerated cellulose when determined as specified in 4.4.

3.4 Moisture permeability.-The moisture permeability of the sheeting shall not exceed 0.090 grams of water per 9.9 square centimeters per 24 hours when tested as specified in 4.5.

3.5 Water immersion.-There shall be no loss of coating from the regenerated cellulose when tested as specified in 4.6. Slight blistering will be permitted.

3.6 High temperature hardness.-The sheeting shall not be tacky or soft at 170°F when tested as specified in 4.7.

3.7 Residue characteristics.-The material shall withstand burning without melting and dropping from the specimen when tested in accordance with 4.8.

3.8 Heat sealing.-The seam strength shall be 1.00 pounds, minimum, per inch of width when tested in accordance with 4.9.

3.9 Reactivity.-The material shall not show harmful effects upon the propellant when subjected to the 90°C vacuum stability test in accordance with 4.10.

3.10 Thickness.-The thickness of the sheeting shall be .0014 ± .0003 inches when determined in accordance with 4.11.

3.11 Uniformity.-The sheeting shall be uniform in color, texture, finish and physical properties.

3.12 Surface finish.-The surface may be smooth and glossy or may be dull and satin like, but shall not be rough or pitted sufficiently to retain particles of dust and dirt when wiped with a dry cloth.

3.13 Workmanship.-The sheeting shall be free from porosity warpage, checks, cracks, blisters and other defects affecting the serviceability of appearance.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.-A lot shall consist of cellophane coated with material from the same batch or blend and applied with no change in operating processes or conditions.

4.2 Sampling.-Ten percent shall be selected but not more than 10 or less than 3 of the bales or boxes comprising the lot. If there are fewer than 3 in the lot, all the bales or boxes shall be sampled. One roll shall be removed from each selected bale or box, at least 50 feet shall be unrolled from the roll, and a sample of two running yards shall be cut from the strip. Each sample shall be labeled so that the bale or box from which it was taken can be identified. A composite sample shall be made of the lot, consisting of approximately 1 yard from each of the primary samples. All samples shall be labeled giving the name of the material, grade, manufacturer, plant, purchase order, and the number of pounds in the lot. All acceptance tests shall be made on the

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composite sample, using as many of the pieces as possible in making the test. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the purchase description. The primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements.

Note: Each of the strips finally selected for testing shall be from a different mill roll.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550

4.4 Bursting strength.-Bursting strength shall be determined in accordance with Method No. 112 of Specification UU-P-31. Before test, the specimen shall be conditioned as specified under determination of tensile strength, Specification L-P-406. A minimum of ten tests shall be conducted and the average value reported to determine compliance with 3.3.

4.5 Moisture permeability.-Specimens of convenient size shall be fastened over the only opening of a cup 1 1/2" in diameter by 1/2" deep containing 5 gm of distilled water. The specimens shall be sealed to the containers all around with a moisture proof seal. The use of a wax, varnish or lacquer to make this seal shall be permissible. The containers shall be placed in a desiccator which contains concentrated sulphuric acid to a depth of 1-1/4 to 1-3/4 inch, and the desiccator shall be placed in an oven operating at 50 $\pm$ 2°C. The container with water and sheeting shall be weighed each 24 hours until the weight loss for two consecutive 24-hour periods shall not vary by more than 20% of the larger. Results shall be expressed as grams of water lost per square centimeter of film per 24 hours to determine compliance with 3.4.

4.6 Water immersion.-Sheeting shall be immersed in distilled water which has been freshly boiled and cooled to 25 $\pm$ 2°C. After 4 days at this temperature, the sheeting shall be examined. While still wet, the sheeting shall be rubbed between the fingers and examined for evidence of delamination to determine compliance with 3.5. Slight blistering will be permitted.

4.7 High temperature hardness.-The tests shall be conducted after the specimens have been subjected to a temperature of 170 $\pm$ 5°F for 1 hour and while the specimens are still at that temperature. Tackiness shall be judged by squeezing the film between the fingers to determine compliance with 3.6. The specimen shall also be scratched with the fingernail while the film rests on a glass or plate and shall be judged satisfactory in hardness if none of the film is scraped off or torn.

4.8 Residue characteristics.-Samples shall be selected by the inspector and tested in accordance with Specification L-P-406, Method 2022. Observe for burning without melting and dropping from the specimen.

4.9 Heat sealing.-Four pieces shall be cut from each sample measuring 10 x 3.5 inches, these shall be folded in half, providing four pieces 5 x 3.5 inches. One of the 5 inch sides of each specimen shall be sealed with a seam $\frac{1}{2}$ inch wide using a temperature of 325 $\pm$ 25°F, a pressure of 40 $\pm$ 5 pounds per square inch and a dwell time of 1.0 $\pm$ 0.5 second. (A model 12A Sentinel Heat Sealer manufactured by Packaging Industries, Montclair, N.J., has been found suitable). Five 1-inch specimens shall be cut from each piece perpendicular to the line of the seal and the resulting 20 specimens shall be cut 1.25 inches long. The seam strength shall be determined with a pendulum-type instrument, or equivalent, provided with a scale ranging from zero to fifteen pounds graduated to one-eighth pound. With the jaws of the instrument set 1 inch apart, 2 1-inch specimens shall be placed together and one end of the pair of specimen shall be placed in the upper jaw and the other end of the pair in the lower jaw. The jaws shall be separated at a speed of 12 inches per minute. The results obtained from the ten pairs of specimens shall be averaged and the average shall be divided by two, and the resulting value shall be reported as the average seam strength per inch of width to determine compliance with 3.8.

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4.10 Reactivity

4.10.1 Approximately 5 grams of M8 Propellant shall be reduced to a particle size of approximately 12 mesh. Approximately .5 gram of the cellophane sample shall be cut into pieces approximately 0.25 inch square. Three heating tubes of the apparatus shown on figure 1 shall be selected to one of the tubes 2.50 ± 0.05 grams of the M8 Propellant shall be added, and to a second tube 0.25 ± 0.05 gram of cellophane shall be added, and to the third tube 2.50 ± 0.05 grams of M8 Propellant and 0.25 ± 0.05 gram of cellophane shall be added. To each tube, a capillary tube shall be connected and the connection shall be sealed by means of 1 ml of mercury. The apparatus shall be clamped so that the long length of the capillary tube is in a nearly vertical position and the lower end rests on a solid support. Seven milliliters of mercury shall be transfered to the cup at the lower end of each capillary tube. By means of a vacuum pump the system shall be evacuated until the pressure is reduced to approximately 5mm of mercury. The pump shall be disconnected and the vertical height of the column of mercury in each of the capillary tubes shall be measured. The respective vertical height of the mercury in the cups shall be measured and subtracted. The difference is designated at h_1. The room temperature (t_1) and the barometric pressure shall be noted. All three heating tubes shall be inserted in a constant temperature bath (Drawing DP-44282) containing a solution of glycerin in water having a specific gravity of approximately 1.05 which will maintain a temperature of 90° ± 0.5°C in the heating tubes. The temperature within the heating tubes shall be determined by means of a calibrated thermometer inserted to the bottom of one of the tubes specified for the test and the assembly shall be completely inserted in one of the bath wells. The necessary exposed mercury column and bore corrections shall be applied to the observed temperature reading. After 40 hours, the heating tubes shall be removed and the tubes shall be allowed to cool to room temperature. Each vertical height of the mercury column in the capillary tubes shall be measured and from each the respective vertical height of the mercury in the cups shall be subtraced. The difference is the value H. The room temperature (t) and the barometric pressure shall be noted. The volume of gas at standard temperature and pressure liberated during each test shall be calculated as follows to determine compliance with 3.9:

$$\text{Volume of gas in ml} = (A/C(B-H))\left(\frac{273P}{760\,(273/t)}\right) - (A/C(B_1-H_1))\left(\frac{273P_1}{760\,(273/t_1)}\right)$$

where:

A = volume of the heating tube minus 5.00 ml (volume of explosive)

B_1 = the total length in mm of the capillary tube minus the vertical height of the column of mercury in the cup before heating

B = the total length in mm of the capillary tube minus the vertical height of the column of mercury in the cup at the end of the test

C = the determined unit capacity of the capillary tubing in ml per mm

P_1 = the barometer reading when the test is started minus H_1

P = the barometer reading at the end of the test minus H

If tests are performed using 2.5 grams of M8 Propellant and 0.25 gram of cellophane, results obtained shall be extrapolated to 5.00 grams M8 Propellant and 0.5 gram cellophane and shall be reported on that basis.

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4.10.2 Calculation.-From the volumes of evolved gas calculated above, the reactivity caused by contact of the sample of cellophane with the M8 Propellant shall be calculated as follows to determine compliance with 3.9:

$$\text{Reactivity} = X - (Y / Z)$$

where:

X = the volume of gas evolved by the mixture of the sample of cellophane and M8 Propellant
Y = the volume of gas evolved by the sample of cellophane alone
Z = the volume of gas evolved by the M8 Propellant alone.

4.10.3 Calibration of glass tubes.-The volume in ml of the 15.5 ml heating tube shall be determined by running mercury from a burette into the tube until it is filled to the level at which the ground glass joint of the capillary tube will make contact with the mercury. Seven milliliter of mercury shall be transferred to the cup at the lower end of the capillary tube, the tube shall be clamped in an upright vertical position and the height in millimeters of the mercury column in the capillary tube shall be measured, these values shall be added, and the height of the mercury column (approximately 25mm) determined as above shall be subtracted. The capacity of the capillary tube per unit of length shall be calculated as follows:

An accurately weighed portion of approximately 10 grams of mercury shall be transferred to the cup at the lower end of the capillary tube. The tube shall be manipulated so that when the tube is horizontal the mercury is contained in a continuous section of the longest part of the tube. The length of the mercury column shall be measured. This operation shall be repeated twice with the mercury in 2 other parts of the longest section of the tube. The three measured lengths of the mercury column shall be averaged and the unit capacity shall be calculated as follows to determine compliance with 3.9:

$$C(\text{ml per mm}) = \frac{W}{13.59L}$$

where:

C = capacity
W = the weight of mercury used
L = the average length in millimeters of the mercury column

4.11 Thickness.-The cellophane shall be measured to assure that the thickness complies with the requirement of 3.10 by use of micrometer or any other suitable instrument.

4.12 Resubmission and retest.-If the composite sample, or any primary sample subjected to tests, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted, provided new samples, selected in accordance with 4.2, pass all the tests required by this specification and that 3 additional primary samples pass any test or tests failed by the material on original submission.

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5. PREPARATION FOR DELIVERY

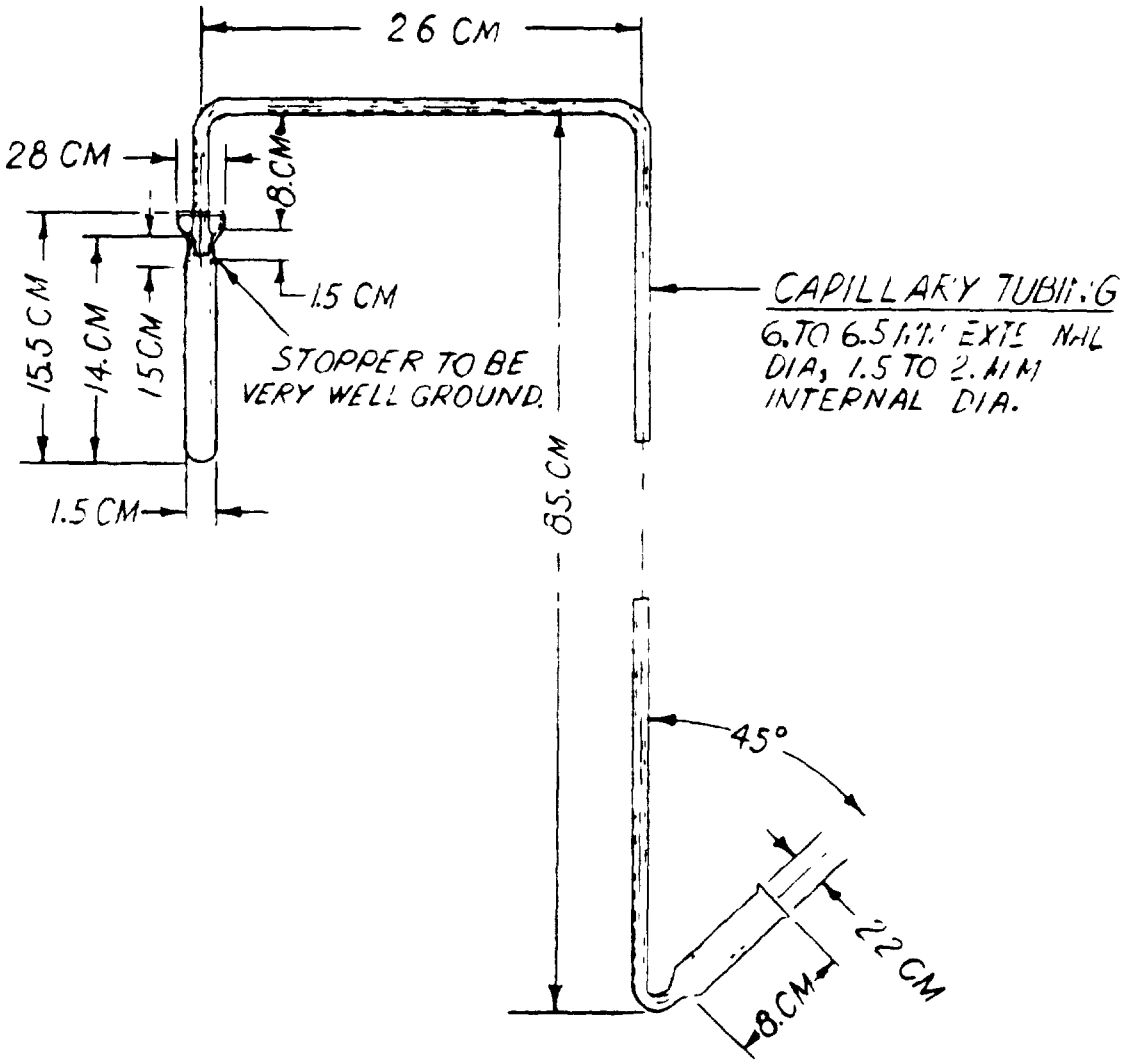
5.1 Packing.-Coated cellophane film shall be furnished in rolls tied with tape and packed in the form of bales or in boxes. Rolls packed in bales or boxes, shall be wrapped in moisture proof paper. Boxes shall be so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery. A bale or box shall contain not more than 200 pounds of film and shall be of sufficiently rigid construction to withstand stacking 10 feet high in storage.

5.2 Marking.-Unless otherwise specified, each shipping container shall be marked in accordance with the applicable requirements of Specification MIL-P-10025.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacturer, use of sell any patented invention that may in any way be related thereto.

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APPARATUS FOR REACTIVITY TEST

FIGURE 1

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